



HTML Essentials

Hypertext Markup Language, or **HTML**, gives a web page its structure and meaning. Browsers parse the HTML elements and build a Document Object Model (DOM) that represents the page's structure.

Basic Page Skeleton

HTML

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport"
          content="width=device-width,
                  initial-scale=1.0">
    <title>Page title</title>
    <link rel="stylesheet"
          href="styles.css">
  </head>
  <body>
    <h1>Main heading</h1>
    <p>Page content.</p>
  </body>
</html>
```

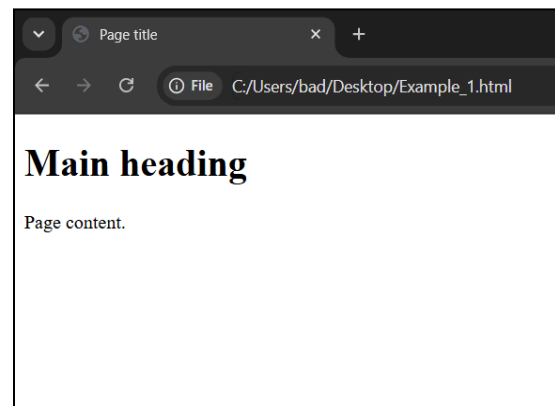


Figure 1. Browser Render of the Basic Page Skeleton

- The `<head>` element contains metadata and linked resources, while the `<body>` element contains visible page content.
- Most elements wrap content and use opening and closing tags. Empty elements—including `<img>`, `<br>`, `<hr>`, `<meta>`, and `<link>`—contain nothing between tags, so they do not use a closing tag.



Elements, Attributes, and Structure

HTML

```
<header><h1>Site title</h1></header>
<main>
  <p class="intro">
    Welcome!
    <a href="about.html">About</a>
  </p>
  
</main>
```

Site title

Welcome! [About](#)



Figure 2. Element Attribute Showcase

- Use semantic elements such as `<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, and `<footer>` when they describe the content.
- Attributes go inside the opening tag as `name="value"`. A class groups elements that share styles. An id names one unique element for CSS, links, or JavaScript.
- Use clear link text. `href` can point to a URL, another file, or an in-page ID beginning with #.
- Give meaningful images concise alt text. Check the filename, folder, extension, and letter case when an image is missing.

Text, Lists, and Tables

HTML

```
<h2>Section heading</h2>
<p>A paragraph of text.</p>
<ul>
  <li>First item</li>
  <li>Second item</li>
</ul>
<table>
  <tr><th>Name</th><th>Role</th></tr>
  <tr><td>Ana</td><td>Dev</td></tr>
</table>
```

Section heading

A paragraph of text.

- First item
- Second item

Name	Role
Ana	Developer

Figure 3. Example for Text, Lists, and Tables

Use headings in order to show hierarchy. Use `<ul>` for unordered items, `<ol>` for a sequence, and `<table>` only for tabular data.



CSS, or **Cascading Style Sheets**, controls presentation. A rule combines a selector with declarations written as `property: value;`.

Connect CSS and Write a Rule

HTML

```
<!-- in <head> element -->
<link rel="stylesheet"
      href="styles.css">
```

CSS

```
/* inside styles.css */
.card {
  color: #17324d;
  background-color: #eaf2f7;
}
```

An external stylesheet is preferred so styles can be reused and the HTML file can stay focused on structure.

Essential Selectors

Selector	What it targets
<code>p</code>	Every <code><p></code> element
<code>.card</code>	Any element with <code>class="card"</code>
<code>#hero</code>	The element with <code>id="hero"</code>
<code>.card a</code>	Links inside elements with <code>class="card"</code>
<code>a:hover</code>	A link while the pointer is over it

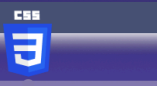
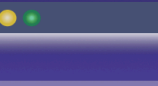
- Use classes for reusable patterns. Reserve IDs for unique page targets.
- Keep selectors short and specific enough to identify the intended elements.



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Cascade and Specificity

- Many text properties—such as `color`, `font-family`, and `line-height`—are inherited from a parent container.
- If the same element has multiple rules, the style displayed will depend on a hierarchy of selectors, with the most specific selector taking precedence. This is known as **specificity**.
 - The hierarchy is as follows: IDs outweigh classes, attributes, and pseudo-classes, and all of which outweigh element selectors.
- If two matching rules have the same specificity, the browser applies the rule it parsed last.

Box Model and Spacing

CSS

```
.card {  
  width: 300px;  
  padding: 16px;  
  border: 1px solid #333;  
  margin: 24px auto;  
}
```

Every element is content surrounded by `padding`, `border`, and `margin`. Padding creates space inside the border; margin creates space outside it. If a box is wider than expected, check `width`, `padding`, and `border` together.

Essential Visual Properties

CSS

```
body {  
  font-family: Arial, sans-serif;  
  font-size: 16px;  
  line-height: 1.5;  
  color: #222;  
  background-color: #fff;  
}  
  
h1 { text-align: center; }  
img { max-width: 100%; }
```

Include units, such as `px`, `rem`, and `em`, where the property requires it, and end `font-family` lists with a generic fallback in case the earlier, defined fonts aren't available.



Flexbox for Simple Layouts

CSS

```
.cards {  
  display: flex;  
  flex-wrap: wrap;  
  gap: 16px;  
  justify-content: space-between;  
  align-items: center;  
}
```

The `flex` container controls its direct children. `row` is the default main axis; `column` makes it vertical. `flex-wrap` lets items move to another line when space is limited.